

KRISHNA SAI VELIDANDA

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Education

International Institute of Information Technology, Bangalore

Jul 2023 – Jul 2028

Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering

CGPA: 3.51/4.0

Technical Skills

Languages: Python, C++, C, Java, JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, FAISS, Transfer Learning, Fine-tuning, RLHF

Research Areas: Natural Language Processing, Computer Vision, Deep Learning, Reinforcement Learning, Optimization, Recommendation Systems

Tools & Platforms: Git, Docker, Google Cloud Platform, Weights & Biases, Linux, Jupyter Notebook

Libraries: NumPy, Pandas, OpenCV, FastAPI, Pymoo, BoTorch, LangChain, Matplotlib

Research Experience

EEG-based Stress and Recovery Analysis

Jan 2026 – Present

Research Project, IIITB | Supervisor: Dr. Sakshi Arora

Bangalore, India

- Designed a multi-stressor protocol (cognitive and emotional tasks) and recorded simultaneous 8-channel EEG (500 Hz) and single-lead ECG to study stress induction and recovery dynamics.
- Extracted band power, spectral entropy, and frontal asymmetry from EEG; HRV metrics (RMSSD, SDNN, LF/HF) from ECG across 4,939 analysis windows; trained Random Forest, XGBoost, SVM, and 1D CNN-LSTM models.
- Achieved 94.4% EEG stress detection and 95.6% combined EEG+ECG accuracy with CNN-LSTM (LOSO CV); showed ECG-only reaches 65.0%, confirming EEG-ECG pathways are largely independent ($r = 0.08$).

Medical Report Generation from Medical Images

Jan 2026 – Apr 2026

Research Project, IIITB | Supervisor: Dr. Sushree Sangeeta Behera

Bangalore, India

- Built an end-to-end inference pipeline for medical report generation from Whole Slide Images using the Virchow tile encoder and PRISM slide aggregator across pathology and radiology datasets (CheXpert, GBM).
- Achieved 0.70 ROUGE-1, 0.46 BLEU, and 0.87 BERTScore F1 on 400 radiology reports; reached 76.8% entity accuracy on molecular markers (ATRX) with CONCH embeddings for zero-shot pathology report generation.

Video Analytics for Human Action Recognition

Jan 2025 – Dec 2025

Research Project, IIITB | Supervisor: Dr. Sushree Sangeeta Behera

Bangalore, India

- Benchmarked 5 deep learning architectures (ViViT, VideoMAE, YOLOv8, ResNet-CRNN, PoseNet) on 40,000+ videos across 3 datasets (UCF-101, UCF-Crime, Kinetics-400) for human action recognition and violence detection.
- Achieved 99.96% accuracy and 0.99 F1-macro with VideoMAE across 101 action classes; attained 96.5% top-1 and 100% top-5 accuracy with YOLOv8 for violence classification.
- Built a real-time skeleton-based recognition pipeline supporting multi-person tracking (up to 5); conducted systematic hyperparameter tuning and identified a 4.3% accuracy gain from dataset scaling (350 to 2000 samples).

Publications

- EEG-based Stress Detection using Multi-Modal Physiological Signals** – manuscript in preparation, targeting journal submission (2026).

Projects

Domain-Aware Layer-Sensitivity Pruning (DALSP) | PyTorch, Transformers

Apr 2026

- Designed an entropy-based adaptive pruning method for Phi-3.5-mini (3.8B params) using Shannon Entropy to modulate per-layer pruning rates; achieved 20% neuron reduction, 0.56 GB VRAM savings, and 49% throughput gain (11.84 to 17.63 TPS) without retraining.
- Generated domain-specialized subnetworks (Math, Code, Law); Math subnet improved GSM8K accuracy from 42% to 53%.

RACS: Risk-Aware Cold-Start Recommendation | PyTorch, FAISS, Llama 3.1

Apr 2026 – May 2026

- Built a cold-start recommendation framework using Llama 3.1 for contrastive item ranking (ACS) and Bayesian uncertainty quantification (SND) with risk-aware safety constraints.
- Achieved 0.68 NDCG and 1.0 Hit Rate on MovieLens-1M ($k=5$); benchmarked against EASE, SASRec, and ItemKNN with multi-seed evaluation and Wilcoxon significance tests.

Multi-Objective Optimization for Image Classification 🧠 | *PyTorch, Pymoo, BoTorch* Apr 2026

- Performed 3-objective CNN architecture search (accuracy, inference time, model size) over a 9-dimensional hyperparameter space on CIFAR-10 and Fashion-MNIST using Gaussian Process surrogates (qNEHVI) and evolutionary search (NSGA-II).
- Identified a 25-point Pareto front with 87.8% peak accuracy; NSGA-II dominated with 19/25 surviving solutions while qNEHVI converged in 3x fewer evaluations (50 vs 150 trials).

FLARE 2.0: In-Network HPC Aggregation 🧠 | *C, Shell, PsPIN, SST* Jan 2026

- Implemented end-to-end in-network allreduce handlers on PsPIN for dense/sparse HPC collectives, validating packet-level aggregation across `ar_single`, `ar_multi`, `ar_tree`, and `ar_single_sparse`.
- Built a reproducible Linux benchmark pipeline using the PsPIN interface, runtime scripts, and packet trace inspection to quantify latency and throughput.
- Extended evaluation with SST simulator support for in-network aggregation, demonstrating system-level benefits of programmable network processing for HPC communication.

Matrix of Truth 🧠 | *FastAPI, Gemini AI SDK, GCP, PyTorch* Sep 2025 – Nov 2025

- Engineered a multimodal misinformation detection platform analyzing text, image, video, and audio using BERT-based NLP, custom deepfake CNNs (90%+ detection accuracy), and Gemini 2.0 Flash for real-time fact-checking.
- Deployed on Google Cloud Run with Docker containerization, Apache Kafka for high-throughput stream processing, and reverse image search via Cloud Vision API; placed 4th among 270,000+ developers at Google Gen AI Exchange Hackathon.

RL for Cloud Debugging 🧠 | *Python, FastAPI, Docker* Apr 2026

- Built an OpenEnv-compatible RL environment with 50 deterministic scenarios across 10 task categories (Dockerfile syntax, GitHub Actions secrets, Kubernetes selector mismatches) with dense reward shaping (+0.3 per fix, efficiency bonuses).
- Designed a multi-step agent loop (observe-think-act, up to 10 steps) with 6 action primitives and deterministic grading; built for Meta/HuggingFace/PyTorch OpenEnv Hackathon.

Mutation-Driven Test Generation 🧠 | *Python, PyTorch, TRL* Apr 2026

- Built a mutation testing RL environment using GRPO to train Qwen2.5-Coder-7B for bug-detecting test generation; zero-shot 7B achieved 0.37 mean reward on best library with 27% of episodes scoring above 0.3.
- Implemented AST validation, no-regression gates, and a safety layer blocking 15 adversarial reward-hacking cases (empty tests, subprocess escapes, vacuous assertions); validated env is non-gameable (heuristic baselines score 0.0).

High-Frequency Trading Simulation 🧠 | *C++, Java* Oct 2024 – Dec 2024

- Built a low-latency C++ order matching engine with price-time priority order book supporting limit and market orders; implemented market-making and statistical arbitrage strategies in a Java-based virtual trading environment.
- Analyzed firm profitability under varying market conditions (volatility, spread, order flow); simulated multi-agent trading with configurable strategy parameters and P&L tracking.

Sellorita 🧠 | *Python, Streamlit, Langchain* Nov 2024

- AI-powered platform automating ad generation and campaign strategy leveraging AI-agents and their tool-calling abilities.
- Orchestrated a set of AI agents built with Langchain to generate different kinds of ads for various platforms and suggest marketing strategies for the product.

Achievements

- **2nd Prize:** TruthTell Hackathon (WAVES Summit 2025) – top 2 of 5,600+ global submissions. [🔗](#)
- **3rd Runner-Up:** Google Gen AI Exchange Hackathon 2025 – 4th among 270,000+ developers. [🔗](#)
- **2nd Prize:** MERNify, MERN Stack Hackathon at Synergy'24 – IIITB's tech fest with 3,500+ participants. [🔗](#)
- **Dean's Merit List:** Academic Year 2023–2024 (cumulative GPA $\bar{\mu}$ = 3.6/4.0). [🔗](#)

Relevant Coursework

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|--------------------------|----------------------------|--------------------------------|
| • Machine Learning | • Multi-Objective ML | • Data Structures & Algorithms |
| • Deep Learning | • Recommendation Systems | • Linear Algebra |
| • NLP with Deep Learning | • Probability & Statistics | • Operating Systems |